

Here is a **clean, exam-oriented answer (simple + well-structured for 5–10 marks)** you can directly copy:

Factors for Choosing the Right IoT Protocol

The selection of a suitable IoT protocol depends on the **type of device** and the **function it performs**. The following factors are important:

1. Device Capability

The processing power, memory, and storage of the device.

- Constrained devices require lightweight protocols (MQTT, CoAP).
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2. Power Consumption

Battery-powered devices need low-energy protocols.

- Low-power protocols increase device lifetime.
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3. Network Bandwidth

Limited bandwidth requires protocols with low data overhead.

- Lightweight protocols are preferred.
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4. Communication Model

Depends on how devices communicate:

- Request–Response → HTTP, CoAP
- Publish–Subscribe → MQTT

5. Reliability Requirement

- High reliability needed → MQTT (QoS levels)
 - Less critical data → CoAP
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6. Latency Requirement

Real-time applications need low latency.

- CoAP and WebSocket are suitable.
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7. Security Requirement

Sensitive data requires secure communication.

- Use HTTPS, MQTT with TLS
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8. Scalability

Large IoT systems require scalable protocols.

- MQTT supports large number of devices.
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9. Data Type and Size

- Small data packets → MQTT, CoAP
 - Large data transfer → HTTP
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10. Application Requirement

Choice depends on application type:

- Smart home → MQTT
- Industrial IoT → AMQP

- Web-based systems → HTTP
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Conclusion

Choosing the right IoT protocol depends on factors like **device capability, power, bandwidth, reliability, and application needs**, ensuring efficient and reliable communication.

If you want, I can also give a **very short 5-mark version** or **easy trick to remember all points in exam**.